
AN EXAMINATION OF OPPORTUNISTIC ACTION WITHIN RESEARCH ALLIANCES: EVIDENCE FROM THE BIOTECHNOLOGY INDUSTRY

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EXECUTIVE SUMMARY

Like a photographer trying to take a perfect picture, an entrepreneur trying to increase the odds of survival must learn very quickly that focus is everything. And what demands an entrepreneur's immediate focus is the development of new products. Entrepreneurial ventures depend on the rapid creation of new products to gain access to early cash flows, create legitimacy, grab early market share, and increase their odds of survival (Schoonhoven, Eisenhardt, and Lymman 1990). However, the increasing costs and complexity of new product development are making it difficult for entrepreneurial ventures to contain the assets needed for successful R&D within their boundaries, forcing them to reach beyond their borders to access resources. Barley et al. (1992) document the use of more than 900 contractual research agreements within the biotechnology industry alone. Recent research has also found a positive relationship between the use of alliances in the R&D process and the rate of new product development (Deeds and Hill 1996; Shan, Walker, and Kogut 1994).

This particular study focuses on the use of relational contracts in the R&D process and extends the prior work on relational contracts to create an explanatory model of the deterrents to opportunism within a relational contract. The article begins with a discussion of the traditional modes of deterring opportunism and of modes of deterring opportunism based on the development of a strong cross-bound-

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ary relationship. From this discussion, hypotheses are derived that relate certain characteristics of the alliance (frequency of communication, strength of contractual deterrents, hostage investments, age of the relationship, etc.) to the level of opportunism within the relationship. These hypotheses are then tested on a sample of 109 research alliances in the biotechnology industry.

We found significant evidence that a strong relationship between the partners serves as a much more effective deterrent to opportunistic action than the creation of hostage investments or contingent claims contracts. In particular, the results for frequency of communication and the background of the firm highlight the importance of the top management team's understanding and involvement in the management of research alliances. The strong results for background congruence indicates the importance of shared expectations and understandings between the partners.

The strong empirical results for the hypothesized U-shaped relationship between age and opportunism provide support for the existence of both a honeymoon period in the relationship and a liability of adolescence among research alliances. The data also suggest that the honeymoon period for an alliance will last about 4.6 years. We also find that the honeymoon period for alliances that are of little importance to the future of the firm is only 4.1 years, and in our sample of alliances that were important to the future of the firm, the honeymoon period extended from 4.6 years to 6.1 years. © 1998 Elsevier Science Inc.

INTRODUCTION

Like a photographer trying to take a perfect picture, an entrepreneur must learn very quickly that focus is everything. And what most entrepreneurs must immediately focus on is the development of new products. Entrepreneurial ventures depend on the rapid creation of new products to gain access to early cash flows, create legitimacy, grab early market share, and increase their odds of survival (Schoonhoven, Eisenhardt, and Lyman 1990). However, the increasing costs and complexity of new product development are making it difficult for entrepreneurial ventures to contain the assets needed for successful R&D within their boundaries, forcing them to reach beyond their borders. Barley et al. (1992) document the use of more than 900 contractual research agreements within the biotechnology industry alone. Recent research has also found a positive relationship between the use of alliances in the R&D process, the rate of new product development (Deeds and Hill 1996; Shan, Walker, and Kogut 1994), and the creation of shareholder wealth (Park and Kim 1997; Zahra and Covin 1995).

However, the complexity and uncertainty that are forcing firms to reach beyond their boundaries also mean that the R&D process does not lend itself well to traditional modes of contracting. Risk sharing, contract enforcement, and moral hazard are particularly problematic in joint development efforts (Tapon 1989). Because the R&D process generally requires investment in co-specialized assets, continuous information exchange and technology transfer relational contracts appear to be a superior alternative to the traditional forms of governance (Bolton, Malmrose, and Ouchi 1994; Teece 1986; Williamson 1991).

A significant body of literature has developed that examines the conditions under which intermediate forms of transacting are adopted (Hennart 1988; Kogut 1988; Ouchi and Bolton 1988; Pisano 1989). Intermediate forms of transacting include strategic alliances, joint ventures, franchising, and other forms of nonstandard contracting (Williamson 1985). These forms have been broadly clustered under the term relational contracts (Bolton, Malmrose, and Ouchi 1994; Provan and Gassenheimer 1994; Ring and Van de Ven 1992; Leblebici and Shalley 1996). Relational contracts provide the parties to the contract neither the full benefits nor the full costs of the extreme forms of governance

(hierarchy or market) (Hennart 1988). However, they do allow for the adjustment of incentives and control structures to meet the needs of the project (Williamson 1991).

This study will develop and test a model of the deterrents to opportunism derived from transaction cost economics (TCE) and relational contracting (RC). The study focuses on the use of relational contracts in the R&D process. We have chosen to study research alliance because recent research makes it clear that relational contracts have become an important and pervasive piece of the R&D process (Barley, Freeman, and Hybels 1992; Deeds and Hill 1996; Hergert and Morris 1988; Shan, Walker, and Kogut 1994). However, although there is a significant body of empirical evidence about the motivations and the conditions that lead an organization to choose an intermediate form of transaction, there has been no detailed examination of the determinants of the level of opportunism perceived by a partner in a relational contract.

This study extends prior work to create an explanatory model of the deterrents to opportunism within a relational contract. The article begins with a discussion of methods of deterring opportunism and a comparison of the TCE and RC perspectives. From this discussion hypotheses are derived that relate certain characteristics of the alliance (frequency of communication, strength of contractual deterrents, hostage investments, age of the relationship, etc.) to the level of opportunism within the relationship. These hypotheses are then tested on a sample of 109 research alliances in the biotechnology industry. Finally, the empirical results are used to compare and contrast the relative effectiveness of the TCE and RC solutions to deterring opportunism.

THEORY

In recent years, TCE has driven a large proportion of the empirical research on strategic alliances and joint ventures (Brockhoff 1992; Gates 1989; Kogut 1989; Parkhe 1993; Pisano 1990). Under the assumptions of TCE, actors are frequently inclined to opportunism: self-interest seeking with guile. TCE does not assume that all actors are opportunistic—simply that it is costly and difficult to distinguish ethical actors from opportunistic actors (Williamson 1975). Therefore, entering into economic transactions in which significant investments in specific assets are placed at risk exposes the parties to the risk of opportunism. The parties to the contract can attempt to ameliorate the risks of opportunism *ex ante* through deterrents such as contingent claims contracts including nonperformance penalties, auditing provisions, lawsuit provisions, etc., through the creation of hostage investments or by bonding their firms through co-specialized assets and bilateral dependence (Teece 1986). However, as discussed previously, risk sharing, contract enforcement, and moral hazard can be particularly problematic in joint development efforts (Tapon 1989). Therefore, the parties to the transaction remain vulnerable to *ex post* opportunism.

Opportunism in a research alliance takes the form of failing to fulfill commitments, expropriating proprietary technology, withholding or distorting information, and misrepresenting one's abilities. Opportunism occurs because one or more of the participants finds it to their advantage to maximize their individual returns at the expense of their partner. Therefore, actions or structures that decrease the potential benefits either partner can derive from opportunistic action will decrease the incidence of opportunism within an alliance. In a general sense, TCE views governance of economic transactions as the process of designing specific mechanisms to support the transaction and deter opportunism (Heide 1994). These mechanisms are formal structures such as contin-

gency claim contracts and equity investments that serve to deter opportunism within an economic transaction by creating ex post costs to opportunistic action that deter opportunism ex ante (Williamson 1985).

In contrast, relational contracting views the economic exchanges within the framework of the historical and social context in which it takes place and views enforcement of obligations as following from the mutuality of the interests of the parties to the exchange (Heide 1994; Hill 1995; Kaufman and Stern 1988). RC builds from McCauley's (1963) work on noncontractual business relations and McNeil's (1978, 1980) development of a formal typology of "discrete" and "relational" contracts. Recent work has extended the ideas of RC in to the area of differences in national institutional structures and transactions costs within national economies (Hill 1995) and the perception of risk by the parties to a relational contract (Nooterboom et al. 1997).

RC assumes that the governance of relational exchanges is best carried out through the use of normative processes in which parties to the exchange adopt certain norms and values (Heide 1994). Relational contracts operate in a bilateral system in which the goals and benefits of the exchange can only be reached through joint accomplishment (Barney and Ouchi 1986), and concern for the long-run health of the relationship serves as a restraint on the opportunistic action of individual parties to the exchange (Heide 1994). Therefore, the healthier the relationship, the stronger the deterrence to opportunism. This study specifically tests this hypothesis by empirically examining the strength of the relationship between the parties as a deterrent to opportunistic action.

This study compares the effectiveness of the views TCE and RC on transaction governance by testing the effectiveness of their proposed deterrents to opportunism on our sample. The model hypothesizes that opportunism within an alliance is deterred by two sets of variables: (1) the ex ante structural deterrents embedded within the alliance in accord with TCE and (2) the strength of the relationship between the partners.

HYPOTHESES

Structural Deterrents

Structural deterrents are the primary mechanisms proposed by TCE to deter opportunistic action. Structural deterrents are used to adjust the incentives of the partners ex ante and to restrict the incentive of a partner to act opportunistically. These mechanisms, created through negotiation and defined in the alliance contract, seek to adjust the payoff structure of the alliance to reward mutual cooperation and penalize opportunism. Incentives for opportunism can be inhibited by contractual safeguards that inflict penalties for noncooperative behavior. These safeguards include protection of trade secrets, reporting and auditing requirements, arbitration clauses, and numerous others. The purpose of these clauses is to provide an ex post deterrent to opportunism. However, as Schelling (1963) notes "the purpose is deterrence ex ante, not revenge ex post." The exercise of ex post deterrents is usually costly to both parties. Therefore, these contractual safeguards serve to adjust the structure of the incentives to deter opportunism.

H1: Increased use of contractual safeguards will decrease perceived opportunism.

Equity investment is the purchase of a percentage of the stock of the firm by the partner. Given the risky nature of new biotechnology firms, the limited scope of their product development process, and the importance of these alliances to their survival, these eq-

uity investments are essentially nonrecoverable investments in the relationship. In addition to providing capital and assets to cash-starved biotechnology firms, these investments serve as hostages. Therefore, equity investments increase the commitment of the partner to the alliance and provide an *ex ante* deterrent to opportunism by reducing the potential benefits from opportunistic action (Schelling 1963; Williamson 1985).

H2: An equity investment by the partner firm will decrease perceived opportunism.

Strength of the Relationship

As discussed earlier, RC argues that concern for the long-term health of the relationship leads to self-monitoring on the part of the parties to the exchange and the deterrence of opportunistic action. Therefore, characteristics of the partners or the relationship that are conducive to the creation of a strong, long-term relationship should serve as deterrents to opportunism. These characteristics include the cultural similarity of the partners, the frequency of communication between the parties, the number of active alliances between the parties, and the age of the relationship. Each of these hypotheses is developed in the following paragraphs.

The idea that cultural differences will affect the perceived level of transaction costs has been developed from the work of Ouchi (1980) and Maitland, Bryson, and Van de Ven (1985). The basic argument regarding the role of culture in determining the perceived level of transaction costs is that norms of reciprocity exist between members of a common culture, inducing parties to refrain from engaging in opportunistic behavior (Jaeger and Bklaiga 1985). Several studies from the field of international management provide evidence of the link between national culture and the perception of transaction costs (Hill 1995; Jaeger and Bklaiga 1985; Kogut and Singh 1988; Shane 1992).

It is not just national culture but organizational culture that influences the perception of opportunism within an alliance. Harrigan (1986) and Parkhe (1991) have both argued that corporate culture homogeneity among partners is even more important to a global strategic alliance's success than similarity in national culture. We extend this argument to the cultural differences between the academic community and the for-profit pharmaceutical and chemical companies.

An area of continual conflict between academic institutions and for-profit organizations has been the publication of research results (Dasgupta and David 1987). Academics desire to publish the results of their research early and often. In contrast, for-profit firms are concerned about expropriation of the benefits of their research and protecting intellectual property and competitive advantage. An examination of the social ethos of scientist versus technologist found that "in the science club, one's discoveries and inventions must be completely disclosed, whereas in the technology club, such findings must not be fully revealed" (Dasgupta and David 1987, p. 528). Under these circumstances, an organization that was established and run by a team with a background in academics should have less conflict and perceive less opportunism in a relationship with a university or biotechnology firm with similar backgrounds.

This decrease in opportunism results from the congruence of the goals and values of the two organizations. A significant body of research has followed Hambrick and Mason's (1984) work on the importance of upper management's background, experience, and demographics in determining organizational strategies and culture (Bantel and Jackson 1989; Scherer and Huh 1992). In the field of biotechnology, the founders

and managers of biotechnology companies generally come from one of two backgrounds: (1) academic or (2) pharmaceutical or chemical industry (Burrill and Lee 1993). Managers with academic experience will expect and accept the academic researchers' demand to publish and may in fact encourage the publication of research by their own organization. Similarly, an organization that is established and run by a team with a background in industry will perceive less opportunism in a relationship with a pharmaceutical company or other for-profit firm that discourages the publication of research, because their values and expectations for the relationship will have been shaped by their industry experience. Therefore, a similarity in the backgrounds of the partners will decrease the perceived level of opportunism.

In addition, membership in these "clubs" or networks deters acts of opportunism in an alliance owing to the risk of social sanctions being imposed by the network. A disparity between the treatment of members of a social network and those who are outsiders is well documented in sociology and anthropology (Amir 1975; Murphy 1956). In fact, surprising levels of trust and mutual forbearance frequently exist within a social network at the same time that high levels of distrust and opportunism exist in dealings with nonmembers of the network (Granovetter 1985). In the pharmaceutical industry, a recent survey of Ph.D. students found that the research reputation of a firm and the ability to publish their research were important criteria in choosing an employer in the pharmaceutical industry. In addition, the publication records of the major pharmaceutical firms were found to be highly correlated with the Ph.D. students rankings of the desirability of the firm as an employer (McMillan and Deeds 1997). The combined effect of shared norms and values and the threat of social sanctions leads to the next hypothesis.

H3: A congruence between the backgrounds of the partners in a research alliance will decrease perceived opportunism.

The idea that contact between social groups decreases conflict is not unique and has a long history in the fields of anthropology, sociology, and studies of race relations (Amir 1969, 1975; Levin 1965; Murphy 1956; Oaker and Brown 1986). At least three casual explanations for the "contact hypothesis" have been presented. The first is that intergroup interaction leads members of different groups to develop positive sentiments for one another, which will attenuate intergroup conflict (Homans 1950). The second explanation is that intergroup contact decreases polarization between the groups by maintaining the permeability of group boundaries, facilitating the transfer of information that contradicts group biases and creating intergroup loyalties that moderate pressure for conformity and radicalism (Ashmore 1970). The third explanation is that contact between the groups allows for the resolution of disputes as they arise, avoiding the accumulation of grievances and the development of grudges. This in turn makes the relationship less political than it is when contacts between groups are infrequent (Patten 1970; Thomas 1976). Our research is unable to distinguish between these rival explanations, but we are quite clearly able to test the impact of contact on the perceived opportunism within the alliance.

H4: As the frequency of contact increases, perceived opportunism will decrease.

Researchers in organizational theory and strategy have discussed the role of organizational or structural attachment between exchange partners. Continued interaction between partners leads to the investment in the development of specialized skills and rou-

tines adapted to the exchange (Blau 1964; Granovetter 1985; Levinthal and Fichman 1988; Seabright, Levinthal, and Fichman 1992). In fact, a venture that is embedded within a web of agreements between the partners is more likely to be stable. This is due to the value of the expected future interactions and the threat of reciprocal opportunism (Kogut 1989). Opportunistic action within one alliance puts all the alliances at risk. Therefore, the greater the number of active alliances between the partners, the less opportunism there will be within a particular alliance. This leads directly to the next hypothesis.

H5: The number of active alliances between the partners will be negatively related to the perceived level of opportunism in the alliance.

As the level of interaction between the parties increases, the investment in co-specialized assets increases and the level of bilateral dependence increases between the parties (Teece 1986). As the relationship between the partners develops over time, two forces are at work that deter opportunism. The first is that the investment in the development of skills and routines adapted to the exchange increases (Blau 1964; Granovetter 1985; Levinthal and Fichman 1988; Seabright, Levinthal, and Fichman 1992). These specialized skills and routines include specific knowledge about the structure and operation of the partner organization, the skills and abilities of the personnel within the partner firms and cross-boundary teams and structures to create, process, and share information. These skills and routines are investments in specific assets adapted to the transaction and are at risk if cooperation between the partners breaks down. The second force working to deter opportunism is the development of social relations among the parties. The creation of social relations increases the threat of damage to personal reputations as well as firm reputation and the desire to maintain the loyalties and friendships established across organizational boundaries (Homans 1950; Granovetter 1985). This leads to the next hypothesis.

H6a: As the experience the partners have with one another increases, perceived opportunism will decrease.

However, the relationship between age and opportunism is unlikely to be as simple as the proposed linear model. We believe that the more appropriate form for the relationship between age and opportunism is an inverted U-shape (Figure 1). An alliance, like any social relationship, is likely to experience an initial honeymoon period. This is due to the fact that an alliance starts with an initial stock of assets, which may include favorable prior beliefs, goodwill, trust, financial investments, or psychological commitment (Fichman and Levinthal 1991). A relational contract, such as a research alliance, must have a certain level of commitment on the part of the actors in order for the relationship to be initiated. Once actors enter a relationship they become bound to the relationship, creating a justification process and a certain amount of behavioral inertia (Salancik 1977). The commitment to the new project and partnership buffers the alliance from early indications of opportunism. Under these conditions, it is more likely that partners will attribute early disagreements to the normal learning process rather than conscious acts of opportunism. In addition, in the early stages of an alliance the potential value of the partnership and the severity of the partners' retaliation to perceived acts of opportunism are uncertain, causing the partners to restrain themselves as they gather information regarding the ultimate value of the alliance and the level of retaliation they can expect for acts of opportunism.

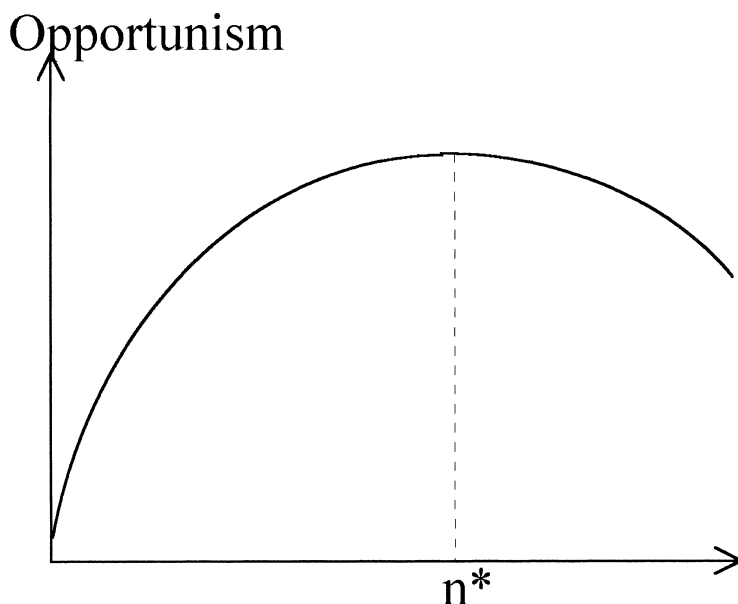


FIGURE 1 Age of the relationship.

Time is an integral part of any complex transaction. Exchange processes are uncertain precisely because they occur over time (Leblebici and Salancik 1982; Pfeffer and Salancik 1978). The duration of a research alliance by its very nature extends the time between initial exchange and evaluation of the outcome, because the outcome and the steps necessary to achieve the outcome are uncertain. In the case of the drug development process, the movement of chemical entities from discovery to formulation to pre-clinical trials to clinical trials and final approval is currently estimated to take 10 years (BIO 1997). By definition, during the initial period before the benefits of the exchange are realized, the relationship is not threatened by any negative outcomes (Fichman and Levinthal 1991). Therefore, it may take years for the parties to a research alliance to gather enough information to begin to estimate the potential benefits from the exchange and evaluate their partner's behavior.

A relational contract is an extended series of exchange interactions, each providing additional information about the character of the partner and the quality of the project, as well as providing the chance for miscommunication or misinterpretation of each others' actions or motives. As in a marriage, experiences are accumulated over the course of the relationship. Therefore, if a firm begins a relationship with a certain amount of favorable beliefs about the partner firm, which is the basis for initiation of the contract, it is likely to take several events for their evaluation of the partner to change (Fichman and Levinthal 1991). In addition, relationships that are plagued by perceptions of opportunism between the partners are likely to dissolve. This selection process should lead to a negative relationship between age and opportunism among the population of alliances at any point in time. Therefore, combining the two effects leads us to conclude that the relationship between age and opportunism is an inverted U-shape with opportunism rising when the alliance is young, then reaching a peak and declining after some initial honeymoon period. This suggests an alternative to H6a.

H6b: There will be an inverted U-shaped relationship between the age of the relationship and perceived opportunism.

Note that H6a and H6b postulate different functional forms for the relationship between the age and opportunism. If the relationship between age and opportunism is nonmonotonic as postulated by H6b, then an important issue is what determines the length of the honeymoon period, or more precisely what determines the point of inflection of the relationship between age and opportunism. As we have argued previously, alliances begin with a certain amount of goodwill or social capital, which may include favorable prior beliefs about their partner and the relationship, goodwill, trust, financial investments, or psychological commitment (Fichman and Levinthal 1991). In general, an alliance will have a longer honeymoon period if the firm is generally positive about the alliance and has made a significant commitment. The strategic importance the firm placed upon the alliance should reflect the firm's attitude and commitment to an alliance. A firm is unlikely to place great strategic importance on an alliance about which it does not have a positive attitude and a significant level of commitment. Commitment to a relationship implies a certain amount of continuity of behavior across time (Fichman and Levinthal 1991). Therefore, there will be a greater level of inertia in the beliefs about the alliance partner, which shields the alliance from a certain level of negative outcomes. Therefore, if the honeymoon is the period of time in which the relationship is shielded from negative outcomes, the length of an alliance's honeymoon period should be positively related to the importance of the alliance to the firm's future. This leads to our next hypothesis.

H7: Alliances that are very important to the firm's future will have a longer honeymoon period.

METHODS

Data Collection

A cross-sectional research design was used to test the proposed model. The cross-sectional design was chosen because of data limitations: no existing databases of research alliances provided the required data. Structured interviews were chosen because of the richness of the data (Kerlinger 1986) and because that method allowed the interviewer to clear up any ambiguity or misunderstanding of the questions. Standard responses and examples were developed by the interviewer to improve the understanding and decrease the ambiguity of the questions. A single interviewer was used throughout the study, which allowed consistent explanations and examples to be provided to all the interviewees.

The top management team was chosen as the target based on information received during the preliminary interviews with industry executives. Therefore, a member of the top management team became the target of the interviews. In particular, the CEO, VP of Business Development, or VP of Research were targeted.

The foci of this study are research alliances. The ideal sample would be a random sample from the total population of research alliances. However, such a random sample is not feasible due to limited information. There is no means of gathering a list of all the research alliances across industries, and in many cases, research collaborations are not made public due to strategic and other concerns. Some degree of external validity was sacrificed, and a sample was chosen judgmentally with the hope that the relation-

ships among the focal variables could be examined while controlling for confounding factors (Cook and Campbell 1979).

The biotechnology industry was chosen for this project because of the widespread use of research alliances within the industry. The rapidly developing technology, the escalating costs of development, and the risks and uncertainty, inherent in the industry, have created a dense network of linkages among biotechnology companies, pharmaceutical companies, and universities. Barley, Freeman, and Hybels (1992) found 3,441 separate linkages among the players in the biotechnology industry with more than 900 (27%) of these agreements covering the research and development of a new product or technology. We defined biotechnology firms as firms established primarily to pursue biotechnological research and development in areas of commercial promise (Barley, Freeman, and Hybels 1992).

To facilitate personal interviews, a geographical subset of biotechnology firms was developed. The areas chosen were Seattle, San Diego, and Philadelphia/New Jersey. *Bioscan* (1992) was used to develop a list of the 132 firms in these areas. All of the target firms were contacted, and interviews were arranged with 57 biotechnology firms. Each firm provided information on up to three distinct alliances. The 43% response rate is higher than previous studies (John 1984; Parkhe 1993). Personal interviews were used to rate 97 alliances, and phone interviews were used to rate 12 alliances. No firm rated more than three alliances.

The high response rate from the top managers of biotechnology firms was encouraging, but it did not eliminate the potential problems posed by a single respondent. Self-reported data from a single source may exhibit systematic variation among the variables due to a defect in the source (Cook and Campbell 1979). These concerns include common method variance, questions about the consistency and reliability of the data, and possible bias due to the social desirability problem. However, reliance on a key informant was the only method available to gather the information.

Reliance on a single source is problematic, however. By limiting targets to the top management team and conducting interviews, some of the potential for misinterpretation of questions and responses was mitigated, and interviews created certainty as to who responded and their knowledge of the alliances. In addition, in response to the threats mentioned earlier, three statistical procedures were used to test for reliability and common method variance.

The first was the use of secondary data to test for convergence (Keats and Hitt 1988). However, in this study, most of the data were not available though published secondary sources. However, there were data available on the history of cooperation between the parties, number of employees, age of the company, and number of alliances in which the company had participated. These data were reported in *Bioscan* (1992). Secondary source data were available for 52 of the 57 firms, but the cooperative history data were limited to 32 firms due to the unwillingness of the other firms to name their partner.

The secondary source data were compared with the interview responses. The number of employees matched in 44 of the 52 cases. The discrepancies in the other cases were attributed to changes during the time between this study's data collection and *Bioscan*'s. The age of the company matched in all 52 cases. The number of alliances in which the firms had participated matched in 38 of the cases. In 10 of the cases, the total number reported in the interview was two or fewer above *Bioscan*'s numbers and was attributable to the time elapsed between *Bioscan*'s data collection and the interviews conducted in this study. In four cases, the interviewee reported a lower number of previous alli-

TABLE 1 Results of Harman Single Factor Test

Variable	Factor #1	Factor #2	Factor #3	Factor #4
Age of the relationship	0.396	0.857	-0.163	0.080
Age of the relationship squared	0.343	0.846	-0.134	0.190
Background congruence	0.093	-0.324	0.127	0.086
Contract strength	0.194	-0.147	-0.550	0.445
Dependence	0.486	0.071	0.250	-0.007
Frequency of communication at the executive level	-0.417	0.390	0.509	0.308
Frequency of communication at the operations level	-0.466	0.426	0.490	-0.103
Importance of the alliance	0.604	-0.240	0.009	-0.072
Opportunism	-0.616	0.174	-0.553	-0.122
Number of active alliances with the partner	0.293	0.346	-0.203	-0.636
Equity investment	0.177	0.001	-0.001	0.612
Performance	0.728	-0.129	0.329	-0.085
Eigenvalue	2.34	2.14	1.38	1.16
% of variance explained	19.5	17.9	11.5	9.7

ances. This may be due to strict adherence to this study's definition of a research alliance or a failure to include or properly account for all of the alliances in which the firm had participated. The largest discrepancy was five alliances, but this amounted to less than 10% of the total alliances in which the firm had participated. The history of cooperation between the parties matched in 31 of the 32 cases and was off by 1 year in the remaining case. Overall, the secondary source data provide fairly strong evidence in support of the consistency and reliability of the data.

The second method applied to test for common method variance was Harman's single factor test (Harman 1967). The assumption underlying the test is that if a significant amount of common method variance exists, a single factor, or a general factor that explains a significant amount of the variance, will emerge from factor analysis when all the variables are entered. The results of the factor analysis are reported in Table 1. The factor analysis reported four factors with eigenvalues above 1.0, which combined explained 58.6% of the variance. The largest single factor explained 19.5% of the variance which is significantly less than a majority of the variance.

Finally, a group of seven interviews were conducted with secondary informants at seven firms. The results of these interviews were compared with the primary informants to determine the interrater reliability. The analyses were computed using Kappa (k) (Fleiss 1981). Kappa is an adjusted correlation coefficient and can be interpreted as a correlation coefficient. The kappas for the individual items ranged from 0.94 to 0.76. The overall kappa was 0.82, which is above the 0.80 level suggested by Landis and Koch (1977). Therefore our data appear to be reliable.

Measures

Measure Development

The interview scripts were developed from a search of the strategy and psychology literature and initial discussions with faculty and doctoral students. Several of the items under consideration were developed specifically for this study, due to the lack of prior

work in this area. This study relied heavily on previous work in developing the questions for the interview and sought to adapt existing measures to the specific context whenever possible. When new items were required, they were developed through discussions with industry executives and business faculty. Initial questions were then tested on a convenience sample of industry executives. After the initial tests, the questions were refined and presented to the industry sample and the business faculty for additional feedback. After several iterations, the final script for the interviews was developed.

The second stage of the measure development was a small pilot study. During this stage, 20 structured interviews were conducted with biotechnology executives, research scientists, technology transfer professionals, government officials, and representatives of industry associations. Responses from these interviews were compared with secondary source material for accuracy and reliability. In addition, interviewees were debriefed and asked for feedback, and their definitions of certain terms were used in the interview questions.

Opportunism

The dependent variable, opportunism, was operationalized as the level of opportunism on the part of the partner that was perceived by the focal firm. A perceptual measure of opportunism was chosen because given the complexity and uniqueness of any specific alliance, objective measures of opportunism are limited. Opportunism is often what a party to the alliance defines it to mean within the context of the specific alliance, and it is the perceived level of opportunism upon which parties to the alliance base their actions. This measure is the dependent variable in this study and is measured by eight questions. These questions are derived from John (1984) and Parkhe (1993) and were modified to fit the specific context. In each of these studies, Cronbach's alpha was above 0.8. The Cronbach's alpha for the opportunism measure in our sample was 0.855. In addition, factor analysis of the eight questions was run. The results provided two factors that had eigenvalues above 1.0. However, the first factor had an eigenvalue of 3.24, explained 54.8% of the variance, and all eight items loaded on factor 1 at 0.62 or above. Given the high level of consistency among the eight factors, the response to these questions was averaged to form the measure of perceived opportunism by the focal firm.

Contractual Safeguards

Contractual safeguards are operationalized by using eight questions to determine the strength of the safeguards included in the contract. Each item asked about the inclusion of a specific deterrent within the contract governing the relationship. These questions were derived from Parkhe (1993) and reviewed by a lawyer familiar with contracts governing research alliances and executives within the field. To measure the relative strength of the safeguards included in the contract, the proportion of the total items included in the contract governing the alliance was used as the measure.

The Partner's Equity Investment

Focal firms were asked to report if the partner firm had made an equity investment in their firm and if so the percentage of the equity owned by the partner. The percentage of the equity owned was used as a measure of a partner's equity investment.

Age of the Relationship

The age of the relationship was measured from the point at which the two firms first began working together on any project. The data were gathered during interviews and verified wherever possible through secondary data, specifically *Bioscan* (1992).

Congruence between the Partners' Backgrounds

During the interviews, informants were asked to discuss the history of the organization including its founding, the backgrounds of the founders, and the backgrounds of the current management team. This information was cross-checked against information listed in *Bioscan* (1992). Firms that were created from academic research and in which a majority of the top management team has experience as academic researchers were classified as having an academic background. Firms were determined to have been created from academic research if one of the founders was a research scientist at an academic institution immediately prior to founding the firm and if there was an equity stake or licensing agreement with the academic institution. A total of 64% (37 out of 57) of the firms in our sample had an academic background. Firms that were spun out of for-profit firms and were founded and run by a majority of executives from the pharmaceutical, chemical, or agricultural industries were classified as having an industry background. A total of 36% (20 out of 57) of our sample had an industry background.

Alliances between firms with an academic background and universities or research institutes were classified as having a congruent background. Alliances between firms with an industry background and pharmaceutical or industrial firms were classified as having a congruent background. Alliances between biotech firms in which both partners had a university background or both partners had an industry background were classified as congruent. All other combinations were classified as noncongruent. A dummy variable was then created, which was coded as 1 if the backgrounds of the partners were congruent and a 0 if the backgrounds of the partners were not congruent.

Frequency of Contact

Respondents were asked how frequently senior executive from the firms communicated and how frequently communication took place at the operational level. Frequency of communication was classified as weekly, monthly, quarterly, yearly, or never.

Number of Alliances with Partner

Respondents were asked how many alliances they were currently engaged in with their partner in the alliance.

Importance of the Alliance

Respondents were asked to rate the importance of the alliance to the future success of their firm on a 5-point scale with 1 = unimportant and 5 = very important. The sample was then split into those alliances that rated as 4 or above on the scale of importance and those that were rated as 3 or below on the scale.

Control Variables

To control for possible confounding factors, several possible control variables were identified and tested. These variables included alliance performance, firm size, dependence of the focal firm on the partner, importance of the alliance, and dummy variables for the type of partner. Only alliance performance and dependence were found to have any explanatory value and were included in the model. A discussion of the rationale for their inclusion and the operationalization of the variables is included below.

Alliance Performance

Traditionally, opportunism has been hypothesized as causing a decline in alliance performance. However, information gathered during numerous interviews with participants in alliances has led us to include performance as a control variable. An alliance that performs below expectations, due to scientific, personnel, or other limitations, creates disappointment and friction between the partners. In many cases, blame for the failure must be assigned leading the focal firm to attribute opportunistic actions to their partner to justify the poor performance.

Alliance performance was rated using three questions. Each measuring a separate aspect of performance: spillover benefits, financial performance, and an evaluation of the overall performance. Because there was no theoretical reason for weighting the responses, an average of the three was used to provide a measure of performance. Cronbach's alpha for the performance measures was 0.7999. The three performance measures were then factor analyzed. The analysis resulted in a single factor with an eigenvalue of 2.15, which explained 71.8% of the variation.

Dependence on the Partner

Parties that are highly dependent may cooperate only due to their limited alternatives. Such a pattern of cooperation by weaker parties can be viewed as compliance rather than cooperation (Heide and Miner 1992). In this case, feelings of insecurity, helplessness, and a lack of control could lead to increased attributions of opportunism. Therefore, it is important that dependence be controlled. Interviewees were asked how long it would take to replace their partner on a scale of less than 1 month, 1 to 6 months, 6 to 12 months, over 1 year, and can not be replaced.

Statistical Modeling Technique

You will recall that we have competing hypotheses, H6a and H6b. H6a predicts a decreasing linear relationship between age and opportunism. H6b predicts an inverted U-shaped relationship with opportunism initially increasing with age and then decreasing after the honeymoon period. We used a quadratic model to test for the existence of a honeymoon. This has the following form:

$$\text{Opportunism} = a + b_1 \cdot \text{Age} + b_2 \cdot \text{Age}^2 + b_3 \cdot Z_1 + b_4 \cdot Z_2 + \dots b_n \cdot Z_n + e$$

If b_1 is positive and b_2 is negative, and both coefficients are statistically significant, there

TABLE 2 Descriptive Statistics

Variable	Mean	SD	Minimum	Maximum
Opportunism	2.32	1.00	1.00	5.00
Contract strength	0.75	0.25	0	1.00
Equity investment	1.65	3.54	0	20.00
Age of the relationship	3.47	2.54	0	12.00
Congruence of background	0.50	0.50	0	1.00
Frequency of contact	2.33	1.05	1.00	5.00
Alliance performance	3.19	0.72	1.67	5.00
Dependence	3.37	1.17	1.00	5.00

is evidence for an inverted U-shaped relationship. However, if b_2 is negative and statistically significant and b_3 is insignificant, then the relationship between age and opportunism matches that of H6a.

To test H7, we split the sample between alliances that were rated as important or very important to the future success of the firm and those that were not. We then ran the regressions again to determine if the relationship between age and opportunism had changed. To control for the possible confounding effects of prior experience with the partner, performance, dependence, etc., on the length of the honeymoon, all of our explanatory variables were included in the models.

RESULTS

Table 2 presents descriptive statistics for the independent and dependent variables. Table 3 presents zero-order correlations for the variables.

Table 4 presents the results from a series of ordinary-least-squares regression analyses of the independent variables on the dependent variable. The base model provides the results for the regression of our control variables, dependence and performance, on opportunism. This model is included to provide a baseline of comparison to the full

TABLE 3 Correlation Matrix

Variable	1	2	3	4	5	6	7	8	9	10
1. Contract	1.00	0.12	-0.08	-0.02	0.06	-0.12	-0.04	-0.05	0.03	-0.10
2. Equity investment	0.12	1.00	0.40	0.39	0.16	0.27	-0.09	0.20	0.04	-0.15
3. Age of the relationship	-0.08	0.40	1.00	0.91	0.11	0.17	0.34	0.23	0.18	-0.11
4. Age of the relationship squared	-0.02	0.39	0.91	1.00	0.07	0.08	0.19	0.32	0.22	-0.09
5. Congruence of backgrounds	0.06	0.16	0.11	0.07	1.00	0.20	-0.05	0.01	0.10	-0.30
6. Contact at executive level	-0.12	0.27	0.17	0.08	0.20	1.00	-0.07	-0.04	0.03	-0.25
7. Number of active alliances with partner	-0.04	-0.09	0.34	0.19	-0.05	-0.07	1.00	0.17	0.09	0.04
8. Performance	-0.05	0.20	0.23	0.32	0.01	-0.04	0.17	1.00	0.24	-0.58
9. Dependence	0.03	0.04	0.18	0.22	0.10	0.03	0.09	0.24	1.00	-0.34
10. Opportunism	-0.10	-0.15	-0.11	-0.09	-0.30	-0.25	0.04	-0.58	-0.34	1.00

TABLE 4 Regression Results—Beta Coefficients

Variable	Base Model	Model #1	Model #2—Low Importance	Model #3—High Importance
Performance	−.526	−.516	−.490	−.532
Dependence	−.217	−.150	−.215	.013
Contract strength		−.070	−.149	.132
Equity investment		−.125	.177	−.013
Background congruence		−.249	−.137	−.370
Frequency of contact		−.288	−.159	−.371
Number of active alliances with partner		.153	.217	.080
Age of relationship		.664	.793	.821
Age of relationship squared		−.752	−.924	−.734
F	32.39	14.57	7.75	6.54
P(f)	0.0001	0.0001	0.0001	0.0001
R2	0.367	0.531	0.525	0.490
n	109	109	55	52

model. The Adjusted R^2 of the base model is 0.367, and the f-statistic is significant at the 0.001 level. Model 1 presents the results for the full model. The adjusted R^2 for this model is 0.531 and a significant improvement in explanatory power over the base model. The coefficient for contract strength is not statistically significant and provides no support for H1. The coefficient for the partner's equity investment is statistically insignificant and provides no support for H2. The coefficient for the congruence between the partners' backgrounds is negative and significant at the 0.001 level. These results provide strong support for H3. The coefficient for frequency of contact between the partners is negative and significant at the 0.001 level. These results provide strong support for H4. The coefficient for number of active alliances between the partners is positive and significant at the 0.1 level. These results provide no support for H5. The coefficient for age is positive and significant at the 0.01 level, and the coefficient of the square of the age is negative and significant at the 0.01 level. These results provide strong support for H6b.

Model 2 presents the results of the regression run on a subsample of alliances that were not rated as being very important to the firm's future. The R^2 for this model is 0.525, and the coefficient for age is significant ($p < 0.01$) and positive. The coefficient for the square of age is significant ($p < 0.01$) and negative. Model 3 presents the results of the regression run on a subsample of alliances that were rated as being important or very important to the firm's future. The R^2 for this model is 0.490, and the significance of the f-statistic is 0.0001. The most interesting result is that the coefficients for both age of the relationship and the square of the age of the relationship are significant at the 0.01 level and have the expected sign. These results alone do not indicate that the importance of the alliance moderates the relationship between age and opportunism with more important alliances having a longer honeymoon period. To assess this, we need to examine the inflection point of each of the models.

The point at which deterrence due to age becomes effective can be found by taking the partial derivatives of our models with respect to age, setting them equal to zero, and solving for the age of the alliance. If we examine the derivative for each of the models, we can see that the honeymoon period does appear to lengthen as the alliance increases in importance.

$$\begin{aligned}\text{Full Sample: } d(\text{Opportunism})/d(\text{Age}) &= 0.2608 - 2(0.0285)\text{Age} = 0 \\ &\rightarrow \text{Age} = 0.2608/0.057 = 4.57 \text{ yr.}\end{aligned}$$

$$\begin{aligned}\text{Low Importance: } d(\text{Opportunism})/d(\text{Age}) &= 0.399 - 2(0.0486)\text{Age} = 0 \\ &\rightarrow \text{Age} = 0.399/0.0972 = 4.10 \text{ yr}\end{aligned}$$

$$\begin{aligned}\text{High Importance: } d(\text{Opportunism})/d(\text{Age}) &= 0.2346 - 2(0.0194)\text{Age} = 0 \\ &\rightarrow \text{Age} = 0.2346/0.0388 = 6.05 \text{ yr.}\end{aligned}$$

These results support H8.

DISCUSSION AND IMPLICATIONS

This study is an initial attempt at understanding the intricacies involved in managing a research alliance. In sum, we have found evidence that a strong relationship between the partners serves as a much more effective deterrent to opportunistic action than the creation of hostage investments or contingent claims contracts. In particular, the results for frequency of communication and the background of the firm highlight the importance of the top management team's understanding and involvement in the management of research alliances. The strong results for background congruence indicates the importance of shared expectations, norms, and values between the partners. This is particularly important in fields such as biotechnology, where there is a clear area of contention between the parties.

Little support is found for the hypothesis that structural deterrents are effective weapons against opportunism. Although the sign for each variable was as expected, the lack of statistical significance for any of the coefficients provides little evidence to support the effectiveness of structural deterrents. One possible explanation for the results on structural deterrents lies in the difference between *ex ante* or expected levels of opportunism and *ex post* or the residual level of opportunism—which is what is measured here. If the level of *ex ante* safeguards chosen were appropriate for the hazard level of the alliance and effective, then the residual level of opportunism would be randomly distributed among the alliances and the level of *ex ante* deterrents would be uncorrelated to the residual opportunism. However, the residual level of opportunism was not particularly low, indicating that even if we assume that the appropriate level of structural deterrents were chosen *ex ante*, there remained a significant amount of *ex post* opportunism. Clearly, significantly more is going on in these relationships than can be controlled by *ex ante* structural remedies such as contracts and hostages. So, Williamson's original contention that under conditions of uncertainty and ambiguity *ex ante* structural deterrents will be ineffective is correct. These types of relational contracts are much more dependent upon the development of a strong cross-boundary relationship between the parties to deter opportunism. These relational contracts require active management and frequent communication between the parties. It is clear that the historical and social context in which the alliance operates will impact the level of opportunism perceived by the parties to the transaction. In short, it is clear that in our sample, the statistical analysis supports RC theory and does not support the structural/contractual deterrents derived from TCE.

The strong empirical results for the hypothesized U-shaped relationship between

age and opportunism provide support for the existence of both a honeymoon period in the relationship and a liability of adolescence among research alliances. The data also suggest that the honeymoon period for an alliance will last about 4.6 years. We also find that the honeymoon period for alliances that are of little importance to the future of the firm decreased to 4.1 years and that in our sample of alliances that were important to the future of the firm, the honeymoon period extended from 4.6 years to 6.1 years.

This statistical results from our analysis of the relationship between age and opportunism have two important implications for future research. First, it highlights the need for longitudinal studies of strategic alliances. Our results provide interesting insights, but our use of cross-sectional data to study an essentially longitudinal problem limits the strength of our findings. There is not only evidence of a relationship between age and opportunism, but of a complex nonlinear relationship that may be moderated by a number of characteristics, including the strategic importance of the alliance, the prior experience of the firm with the partner and other alliances, and the reputation of the partner. In addition, the nature of the alliance undertaking will probably have a significant impact on the length of the honeymoon period. Research alliances in biotechnology have a high degree of uncertainty, and evidence of success or failure builds slowly over a long period of time, which should increase the length of the honeymoon. Additional research needs to be undertaken to study the determinants of the length of the honeymoon.

Second, our results indicate that researchers studying relational contracts, such as alliances, joint ventures, and buyer-supplier relations, must take into account the age of the relationship. There is an interesting dichotomy in organizational research. Research in the organizational ecology tradition explicitly considers age in hazard rate models, but this tradition never examines the performance of individual alliances. Research in the strategic management tradition considers performance, but ignores the role of age in determining performance. Future researchers need to carefully consider the impact age may have on perceptions of performance and opportunism within an alliance.

Our study is subject to several potential biases. In particular, the problem of percept-percept common method variance could lead to overstatement of the statistical relationship among our variables. In addition, the problem of multicollinearity created by the high correlations between the two frequency of communication measures and the high correlation between age of the relationship and the square of the age of the relationship has probably led to an overstatement of the adjusted R^2 's for our regression results. Finally, we must also acknowledge that our focus on biotechnology raises questions about the generalizability of our study beyond this industry. Biotechnology has several unique characteristics, including a long product development and approval cycle, heavy reliance upon often arcane basic scientific research, and a very expensive product development process. However, given these unique characteristics, we still believe that our results are generalizable beyond the biotechnology industry. Research alliances are becoming increasingly important as the level of complexity and interdisciplinarity continues to increase (Mowery and Rosenberg 1989) in the R&D process. As firms continue to reach beyond their boundaries for complimentary assets, deterring opportunism will become an increasingly important topic.

Finally, there are important implications for the practice of management from our results. The first is that the background of the partners matters. Although managers

may not always be able to select a partner with a congruent background, they should understand that differences in background will increase the likelihood of opportunism between the partners and that they should take appropriate measures to minimize the problems created by cultural distance. These measures can include frequent signaling of intentions, the institution of rules and procedures that provide a predictable structure and the basis for performance evaluation, and the development of clear expectations through honest communications (Larson 1991). In addition, our results highlight the importance of frequent contact between the parties. In business relationships, as in other social relationships, the contact hypothesis appears to hold true. If managers desire to make their alliances succeed, frequent contact at the executive level is important. Finally, managers need to understand what appears to be the life-cycle of an alliance. The initial period in an alliance's life will be characterized by low levels of opportunism. However, during the middle years, significantly higher levels of opportunism are likely to be perceived as the project progresses. Finally, after this period the level of opportunism begins to decrease. The length of these periods will be influenced by the importance of the relationship to the parties' future, with the honeymoon period increasing with the commitment to the relationship. If this process is understood and expected by a manager, significant problems can be headed off before they destroy cooperation and derail the alliance.

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APPENDIX A Interview Script

1. When did your firm begin its relationship with this partner?
 2. How frequently are the senior executives from your firm and the partner in contact?
 1. weekly
 2. monthly
 3. quarterly
 4. yearly
 5. never
 3. How frequently does contact between your firm and the partner firm take place at the operational level?
 1. weekly
 2. monthly
 3. quarterly
 4. yearly
 5. never
 4. Evaluate the following statements
 - A. My partner has always provided us a completely truthful picture of their abilities.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - B. My partner has made commitments that they are unable or unwilling to fulfill.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - C. Sometimes my partner alters the facts in order to get what they need.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - D. My partner carries out their duties even if we do not check up on them.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - E. My partner sometimes fails to do things they have promised to do.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - F. My partner seems to feel that it is all right to use my proprietary technology and know-how for their own internal projects without my permission.
 Strongly disagree 1—2—3—4—5 Strongly agree
 - G. My partner seems to feel that it is all right to do anything within their means that will help further their firm's interests.
 Strongly disagree 1—2—3—4—5 Strongly agree
 5. Many alliances result in "spillover effects" for their parent firms. For example a positive spillover effect occurs when knowledge that can be applied profitably to other products is created during alliance activities. An example of negative spillover effects is the undesired transfer of a valuable skill or technology to an alliance partner. In the present alliance, the spillover effects for your firm are:
 1. strongly positive
 2. positive
 3. negligible
 4. negative
 5. strongly negative
 6. Considering the most relevant measure of financial performance for this alliance (return on investment, burn rate, sales, etc.), the financial performance of this alliance in comparison to your expectation is:
 1. far better
 2. better
 3. as expected
 4. worse
 5. far worse
 7. What is your overall assessment of the alliance's performance compared with your expectations?
 1. far better
 2. better
 3. as expected
 4. worse
 5. far worse
 8. Has your partner made an equity investment in your firm?
 If yes, what percentage of your equity does your partner own?
 9. Which safeguards listed below are included in the current formal agreement of this alliance?
 - A. Periodic written reports of all relevant transactions
 - B. Prompt written notice of any departures from the agreement
 - C. The right to examine and audit all relevant records by a CPA firm.
 - D. Designation of certain information as proprietary and subject to confidentiality provisions of the contract.
 - E. Non-use of proprietary information even after termination of the agreement.
 - F. Termination of the agreement
 - G. Arbitration clauses.
 - H. Lawsuit provisions
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